

Day - 2

Python Introduction

Compiler	Interpreter
A compiler translate the compiler source program in a single line	An interpreter translate the source program line by line
It's faster.	It's slower

Programming Files

- * Script .Py
- * data_analysis .Pynb
- * Program .exe
- * document .docx
- * Spreadsheet .xlsx
- * image .png
- * archives .zip
- * Styles .css

Python Indentation

Example

```
if 5 > 2:  
    print ("Five is greater than two!")
```

Example (Syntax Error):

```
if 5 > 2:  
print ("Five is greater than two!")
```

* print () - function (Note book) ✓

You can use the print () function as many times as you want. Each call print text on a new line by default.

Example.

```
print ("Hello world!")
print ("I am learning Python.")
print ("It is awesome!")
```

* Python

Text must be inside quotes.

Either

"double"	or	'single' quotes
↓		↓
"Hello"		'hello'

* Mix text and Numbers

Example

```
print ("I am", 35, "years old.")
```

* Comment Start with

Example

```
# This is a demo code
→ print (13)
→ 13 ✓
```

```
This is a demo
print (13)
(error)
```


Comment

```
# This is a Comment  
print ("Hello, world")
```

```
# print ("Hello, world!")  
print ("Cheers, mate!")
```

Variable Names

A variable can have a short name (like `x` & `y`) or a more descriptive name (age, carname, total_volume).

Rules for Python variables:

- A variable name must start with a letter or the underscore character
- A variable name cannot start with a number.
- A variable name can only contain alpha-numeric characters and underscores (A-z, 0-9, and -)
- Variable names are case-sensitive (age, Age and AGE are three different variables)
- A variable name cannot be any of the → Python keywords.

Example

num 1 = 3	num 3 = 6
num 2 = 4	num 4 = 8

* Variables name are Case-Sensitive ✓

- age = 2
- Age = 4
- AGE = 6

• print (age)
2

• print (Age)
4

• print (AGE)
6

Example

Legal variable names:

my var = "Anso"
my_var = "Anso"
_my_var = "Anso"
myVar = "Anso"
myVAR = "Anso"
my var 2 = "Anso"

Example

Illegal variable

2 my var = "Anso"
my-var = "Anso"
my var = "Anso"

* Many Values to Multiple Variables ✓

Python allows you to assign values to multiple variables in one line:

Example

`x, y, z = "Orange", "Banana", "Cherry"`

`print(x)` → Orange

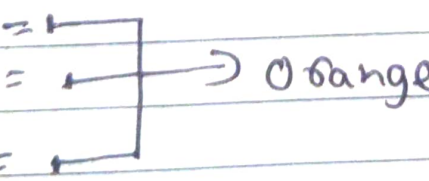
`print(y)` → Banana

`print(z)` → Cherry

* One Value to multiple Variable ✓
And you can assign the same value to multiple variables in one line:

Example

`x = y = z = "Orange"`

`print(x)` =
`print(y)` =
`print(z)` =
 

*

Python Data Types

Data Types	Description	Example	Type
Str abc	Text types: Sequence of unicode characters	x = "Hello world"	Str
int 123	Integer types: whole number	x = 20	int
Float 3.14	Floating points types: decimal numbers	x = 20.5	float
Complex 1+2j	complex numbers with real and imaginary part	x = 1j	complex

Type Casting

```
{
  x = int(1)      # x will be 1
  y = int(2.8)    # y will be 2
  z = int("3")    # z will be 3
}
```

int

```
{
  x = float(1)    # x will be 1.0
  y = float(2.8)  # y will be 2.8
  z = float("3")  # z will be 3.0
  w = float("4.2") # w will be 4.2
}
```

float

```
{
  x = str(1)      # x will be "1"
  y = str(2.8)    # y will be "2.8"
  z = str("3")    # z will be "3"
}
```

str

Multiline Strings

You can assign a multiline string to a variable by using three quotes:

Example

You can use three double quotes:

```
a = """Helloworld!,
I love python,
python is fun."""
```

```
print(a) ... Helloworld!,
              I love python
              python is fun.
```